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World cup 2026 Trends Monitoring

Social Media Analytics

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1 Introduction

The rise of global digital platforms has transformed the FIFA World Cup into a multi-dimensional narrative experience, extending far beyond the pitch into the digital comment sections of platforms like YouTube,. As one of the world's most significant sporting events, the World Cup generates an unprecedented volume of fan discourse, yet this communication is increasingly fragmented across diverse content types and localized discussion spaces. The combination of Youtube's ranking algorithms and the sheer scale of user participation, has turned comment sections into complex ecosystems where national pride, tactical debates, and intense controversies intersect.

The study of YouTube discourse remains a significant methodological challenge due to the platform-mediated nature of content discovery, the brevity of social media text, and the complex participation structures of global audiences. Digital fans do not merely react in isolation; they cluster into specific narrative communities that revolve around distinct anchors, such as match highlights, referee disputes, or star-player legacies. The nuances of these interactions—often characterized by sarcasm, football-specific slang, and high emotional intensity—require a triangulated approach combining network construction with deep content analysis to move from individual comments to meaningful participation structures.

Given the structural complexity of these digital spaces and the critical need for more granular insights into how fans self-organize around global events, this study investigates three fundamental research questions:

1. *Do World Cup 2026 YouTube discussions form participation communities around specific teams, matches, or broader narratives?*

2. *Which narratives dominate detected communities, and how are they expressed through entities, keywords, sentiment, and emotions?*

3. *Which users, videos, matches, teams, players, or controversy-related entities bridge different narrative communities?*

The first research question addresses the structural formation of digital participation communities. By constructing user-user co-commenting networks and applying the Louvain algorithm to a weighted graph, the study evaluates whether repeat commenters naturally cluster into distinct discussion spaces based on shared participation. Community robustness is assessed through modularity measures and size distributions to determine whether World Cup 2026 YouTube discourse is structured into stable narrative groups or remains a fragmented collection of individual reactions.

The second research question adopts a granular content analysis perspective to interpret the meaning behind these structural clusters. Rather than treating fan reactions as monolithic, the study utilizes TF-IDF term weighting, normalized entity extraction, and lexicon-based sentiment and emotion analysis to identify the dominant narratives within each community. This analysis aims to understand how specific anchors —such as tactical blame, referee controversies, or star-player legacies— are expressed through distinctive vocabulary and emotional tones, transforming structural groupings into interpretable narrative communities.

The third research question investigates the bridging roles played by specific users, videos, and entities in connecting these communities. Using betweenness centrality to identify structural brokers and entropy-based metrics to detect cross-community videos and entities, the study explores which elements facilitate discourse across separate narrative boundaries. This step reveals whether certain high-stakes events, such as match eliminations or disputed VAR decisions, act as shared focal points that unify an otherwise divided digital landscape.

By combining network-based community detection with deep content profiling and bridge analysis, this project provides a comprehensive examination of fan participation during the ongoing World Cup. This approach balances structural mapping with qualitative interpretability, offering new insights into how global audiences self-organize around major sporting narratives in platform-mediated environments.

2 Dataset

This study utilizes a custom-collected dataset of YouTube comments concerning the FIFA World Cup 2026, which has been frozen to ensure the reproducibility of the subsequent narrative and network analyses. The final processed corpus consists of 19,152 comments from 134 distinct videos, involving 16,505 unique pseudonymized users.

This dataset was designed to move beyond generic sentiment analysis by capturing a balanced range of discussion spaces, including match highlights, controversies, and tactical debates.

The data collection was performed through a systematic three-step pipeline: candidate discovery via targeted query groups, automatic eligibility filtering, and capped comment collection. During the discovery phase, videos were identified using predefined query groups from a configuration file rather than a single broad search term. To maintain data quality and ensure sufficient density for network analysis, an eligibility filter was applied to exclude videos with fewer than 100 comments, videos shorter than 90 seconds, and content that was non-public, live, or clearly non-English. Furthermore, videos about gaming, simulations, betting and similar content and videos without key metadata were also excluded. For the eligible videos, the collection was capped at 200 top-level comments per video to prevent high-visibility content from disproportionately dominating the sample.

The dataset was constructed by aggregating videos from several distinct narrative contexts, each contributing different discussion characteristics to the corpus:

- **Match Highlights**: This category captures broad, high-visibility discussions that attract a general tournament audience, providing a baseline for participation patterns across the various matches
- **Post-Match Reactions and Star Players**: These videos focus on punditry, fan media, and debates surrounding major figures such as Cristiano Ronaldo or Lionel Messi. This source is critical for capturing narratives related to individual legacies, player-specific praise, or intense fan debates.
- **VAR and Referee Controversies**:This group targets videos explicitly discussing disputed calls, penalties, and offside decisions. These contexts often trigger emotionally intense interactions and serve as a primary source for identifying narratives of injustice or official incompetence.
- **Upsets and Underdogs**: These queries identify unexpected results and surprise performances by smaller teams. Discussions in this category are characterized by narratives of national pride, surprise, or collective disappointment.
- **Tactical Analysis and Blame**: This source focuses on manager criticism, formation analysis, and rivalry narratives. It is particularly useful for identifying communities organized around technical critique and the attribution of blame for team failures.

2.1 Preprocessing

To prepare the raw data for analysis, a targeted preprocessing workflow was implemented to ensure compatibility with the English-oriented text-analysis tools used in the subsequent chapters. Before applying any cleaning, the dataset underwent a series of diagnostic checks to verify video and comment counts, identify duplicate IDs, and assess user overlap across different videos.

The core cleaning process was intentionally light to preserve the natural structure of social media discourse. Raw comments were first deduplicated by their unique comment identifier. Textual noise was addressed by decoding HTML entities, removing URLs, replacing line breaks, and collapsing repeated whitespace. Crucially, punctuation, capitalization, and negation were kept in their original form, as these elements provide meaningful signals for accurate sentiment detection using the VADER lexicon. The project deliberately avoided lemmatization or stop-word removal at this stage to maintain the linguistic nuances necessary for emotional and narrative profiling.

To refine the corpus, two primary filters were applied. A length filter was used to remove comments shorter than 15 characters, which typically provide limited semantic information. Additionally, a rule-based language filter excluded comments identified as likely non-English based on ASCII ratios, common English function words, and football-specific terms. This step ensured that the data would be compatible with the English models of spaCy, NRCLex, and TF-IDF.

Finally, to protect user privacy while preserving the ability to track repeat participation for network analysis, each YouTube channel was mapped to a stable, hashed author_id. This allowed the study to link a single user's activity across multiple videos without relying on raw usernames. The processed comments were then merged with compact video metadata via the video_id, resulting in a finalized dataset of 19,152 comments and 16,505 unique users across 134 videos.

3 Network construction

The network construction phase of this study translates raw participation data into a structural representation of digital discourse. This process involves two distinct stages: the creation of an initial bipartite user-video network and its subsequent projection into a user-user co-commenting network.

3.1 User-Video Network

The first structure is a bipartite user-video network, where nodes represent either pseudonymized users (hashed author_ids) or videos (video_ids). In this graph, an edge is formed whenever a user leaves a comment on a specific video. This representation treats each video as a discrete discussion space and captures the direct participation of individuals across the sampled tournament content. From this network the distribution of the user's partecipation can be observed.

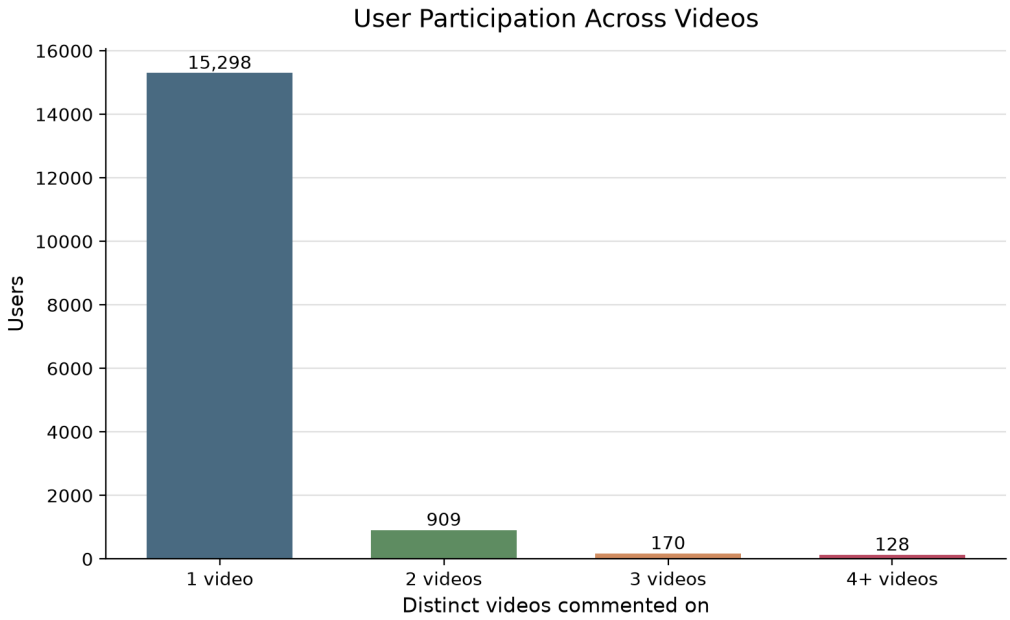


Figure 1: User participation distribution

This bipartite structure serves as the foundation for identifying shared presence in narrative spaces before any user-to-user links are inferred.

3.2 User-User co-commenting Network

The second stage involves projecting bipartite graph built in the first stage into a user-user co-commenting network. In this projected network, nodes represent users only, and a link is established between two users if they have commented on at least one video in common. The strength of these links is captured through edge weights, where the weight corresponds to the exact number of videos shared between two users. For example, a weight of one indicates a single shared discussion, while a weight of three indicates that two users have appeared together in three different video comment sections. It is important to emphasize that this network captures shared participation rather than direct interpersonal ties, such as friendships, follows, or reply-based conversations.

3.3 Edge weight threshold selection

To ensure the structural integrity of the analysis and reduce noise from one-off overlaps, the study evaluated multiple edge-weight thresholds. A threshold of weight ≥ 1 proved to be too dense and noisy, retaining over 1.3 million edges dominated by accidental overlaps on high-visibility videos. Conversely, a threshold of weight ≥ 3

was found to be too restrictive, significantly reducing the network size and potentially fragmenting meaningful narrative clusters.

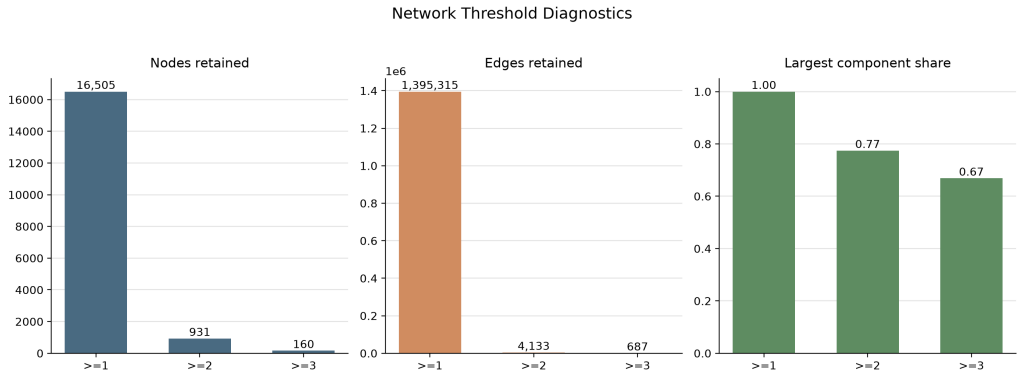


Figure 2: Network Threshold Diagnostics

Consequently, a threshold of weight ≥ 2 was selected for the main analysis. This methodological choice requires users to share at least two discussion spaces to be connected, effectively filtering out weak or coincidental links while preserving a robust backbone for community detection. The resulting main graph consists of 931 users connected by 4,133 repeated co-commenting edges. While the initial preprocessed dataset included over 16,000 users, this thresholding ensures that the final network focuses on a subset of participants whose repeated presence across different videos indicates a more stable form of engagement with the tournament's digital ecosystem. This finalized network provides the structural basis for the subsequent detection of participation communities and the identification of bridge actors within the World Cup 2026 YouTube landscape.

4 Community Detection

The objective of the community detection phase is to identify groups of users who are more densely connected to one another through shared participation than to the rest of the network, in order to answer the first research question. These groups are interpreted as participation communities, representing clusters of users who repeatedly appear in similar World Cup 2026 discussion spaces. This process moves the analysis from individual links to broader structural patterns of digital engagement.

4.1 Methodology

The methodology utilizes the Louvain algorithm applied to the weighted user-user co-commenting network. This specific algorithm is chosen because it searches for partitions where internal connections are significantly denser than external ones. To ensure the results reflect meaningful engagement, the algorithm operates on the main graph thresholded at an edge weight of two or more, meaning users must share at least two different video discussion spaces to be connected. Edge weights are preserved during this step, ensuring that stronger repeated co-participation has a greater influence on the community structure than weaker links.

4.2 Results

The resulting structure reveals a clear partition with a modularity score of 0.5505. This score suggests a meaningful community architecture rather than a random grouping of users. While the algorithm detected 71 communities in total, the distribution follows a sharp long-tail pattern.

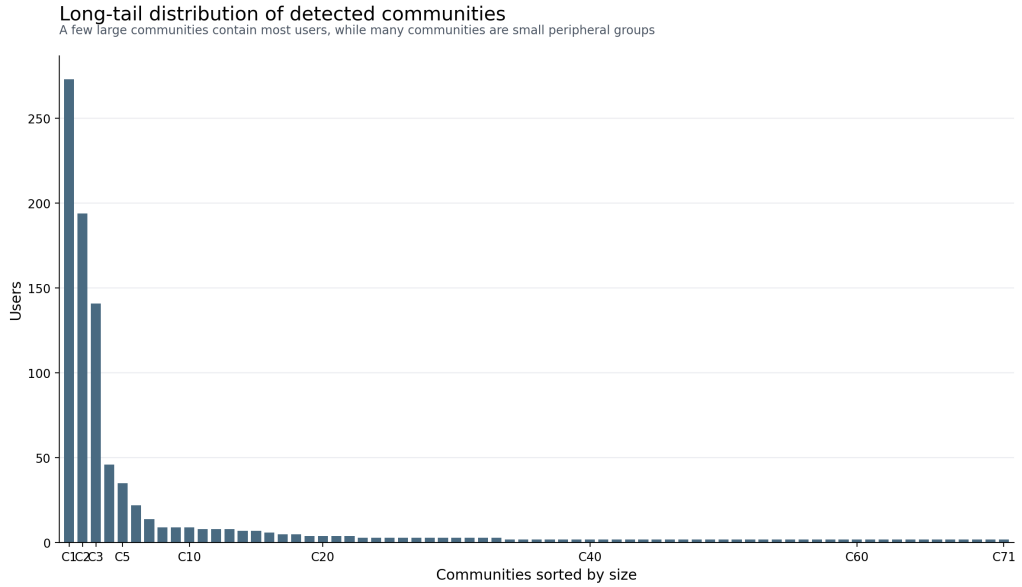


Figure 3: Community size distribution

Most users are concentrated in a few large, interpretable communities, while the majority of communities are small and peripheral. Specifically, the six largest communities, labeled C1 through C6 by size, account for over 76 percent of the users in the graph.

Before incorporating text analysis, these communities were profiled using video metadata, such as sampling categories and context types. The context types represent a set of categorical labels assigned to each video in the dataset to describe its primary narrative environment. These labels were generated using a rule-based heuristic that analyzed video titles, descriptions, and search categories to categorize content into groups such as match highlights, post-match reactions, VAR and referee controversies, star-player legacies, and tactical analysis. This preliminary step allows for an initial understanding of the narrative anchors for each group.

- **C1**: The largest community, is primarily associated with match highlights, representing a broad audience focused on general tournament events.
- **C2**: It is centered around post-match reactions and star-player legacies, particularly debates involving high-profile players.
- **C3**: It is very clearly defined by VAR and referee controversies, specifically those related to match incidents like the Portugal versus Croatia offside dispute.
- **C4**: It appears as a more mixed discussion space involving Germany-England rivalries and VAR issues.
- **C5**: It is uniquely focused on tactical analysis and blame, specifically centered on manager criticism and team performance, such as the discourse surrounding Scotland and Steve Clarke.
- **C6**: Finally, this last big community represents another controversy-oriented community, likely focused on a different set of refereeing or goalkeeping incidents than C3.

The analysis is supported by force-directed network visualizations that illustrate the separation and overlap between these groups. In these visualizations, nodes represent users and colors represent their assigned communities, highlighting how the core of the network is organized around these primary narrative hubs.

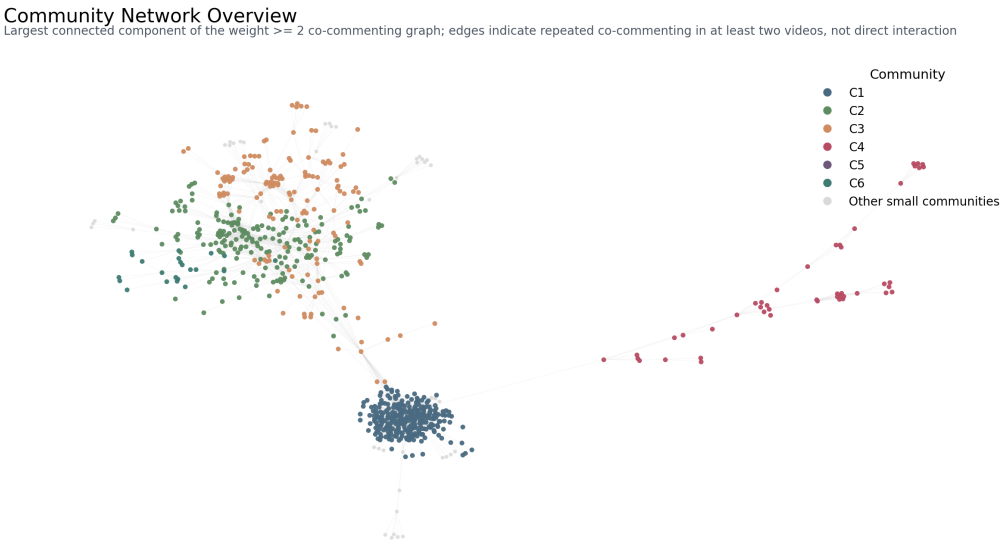


Figure 4: Main Communities

It is critical to note that these are participation communities, not social groups based on friendship or agreement. Membership in a community indicates that users occupy the same digital spaces, regardless of whether they agree or disagree with one another. This systematic approach to community detection provides a structural answer to the first research question by demonstrating that World Cup YouTube discourse is not a monolithic stream of comments but a structured ecosystem. By focusing on the largest communities, the study ensures that the subsequent content analysis is grounded in the most robust and representative participation patterns within the dataset. Significant attention is paid to the methodological caveats, acknowledging that the results are sensitive to the chosen edge threshold and that the heuristic nature of the Louvain algorithm means these partitions represent one useful way to view the data structure rather than an absolute social reality.

5 Content Analysis

The content analysis phase of this study profiles the detected participation communities by examining the actual text of the comments. While community detection reveals how users cluster structurally through repeated co-commenting, content analysis explains the substance of their discussions. This stage directly addresses the second research question, which investigates the dominant narratives within these communities and how they are expressed through entities, keywords, sentiment, and emotions. By moving from structural patterns to thematic evidence, the analysis aims to identify the specific actors, controversies, and tones that define each community, focusing primarily on the six largest clusters that represent the core of the network.

5.1 Input Data

The analysis is performed on a combined dataset that links processed comment text with user community assignments. Each comment inherits the community membership of its author, allowing the study to profile shared discussion spaces rather than individual user interactions. The input data includes the cleaned text of the comments alongside pseudonymized author identifiers to ensure privacy while maintaining the link between a user's structural position and their linguistic output. By focusing on the community-assigned subset of the data, the analysis ensures that the findings are grounded in the established participation structure of the World Cup 2026 YouTube landscape.

5.2 Methology, Techniques and Tools

Several natural language processing techniques are employed to extract narrative signals from the corpus. TF-IDF term weighting is used to identify distinctive unigrams and bigrams for each community, highlighting vocabulary

that is uniquely characteristic of a group rather than common to the entire tournament. Using both unigrams and bigrams is important because some narratives appear most clearly as short phrases rather than single words. Examples include expressions such as offside decision, Steve Clarke, or World Cup. These terms help identify the vocabulary around a community, but they are not treated as full topics by themselves.

To identify the key subjects of discussion, a hybrid entity extraction approach combines general-purpose named entity recognition,spaCy, with a domain-specific football dictionary. This allows for the identification of players and teams alongside critical narrative objects like VAR, referee, or penalty. Furthermore, entity normalization maps multiple surface forms, such as CR7 or Ronaldo, to canonical labels like Cristiano Ronaldo, ensuring that different ways of mentioning the same figure are counted together. This normalization is inspired by NEL/NED because surface forms are mapped to canonical entities. However, it is not full external entity linking. The project does not use Wikidata, DBpedia, or other external entity IDs. That choice is deliberate because the aim is community-level narrative profiling, not the construction of a knowledge-base-linked entity graph.

The emotional and tonal dimension of the discourse is captured through automated sentiment and emotion analysis. Sentiment is measured using the VADER lexicon, which is specifically designed for social media text and accounts for punctuation, capitalization, and negation to determine the polarity of a comment. This is complemented by NRCLex, which scores comments across eight basic emotion categories, those proper to the Plutchik's wheel of emotions, to provide a more nuanced affective profile than simple positive or negative labels.

Finally, the quantitative findings are validated through the manual inspection of representative comments selected for each community. Their main role is to help validate whether the automatically suggested community labels and narrative cues are plausible when checked against real comment text. By triangulating distinctive terms, normalized entities, sentiment scores, and qualitative examples, the content analysis aims to provide a comprehensive interpretation of the narrative anchors that organize digital fan participation.

5.3 Insights

The content analysis reveals that the participation communities are organized around distinct narrative anchors, moving beyond simple sentiment to reveal specific thematic centers for each group.

5.3.1 Top terms related Insights

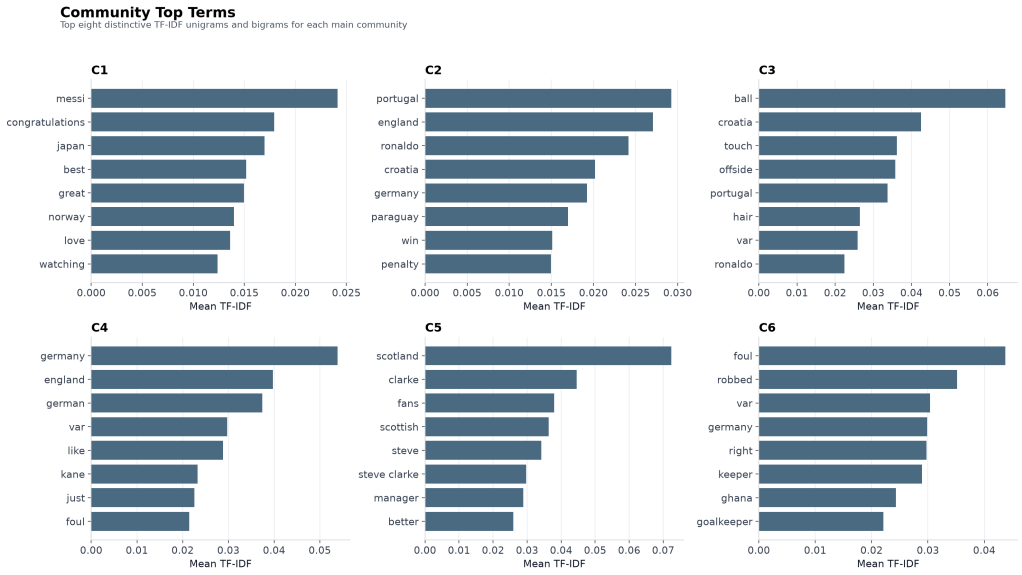


Figure 5: Top Terms per Main Community

The largest community, C1, functions as a celebratory hub for match highlights and player appreciation, characterized by positive terms such as congratulations, best, and great, with discussions often revolving around general tournament success and figures like Lionel Messi.

In contrast, C2 emerges as a space for high-profile post-match debate and star-player legacies, focusing heavily on teams like Portugal and England and players like Cristiano Ronaldo, often mixing admiration with arguments over match fairness.

C3 is one of the most clearly defined communities, centered almost exclusively on a specific VAR and offside controversy during the Portugal versus Croatia match, where users engage in granular debates over technical details like whether a player touched the ball.

C4 follows a similar controversy-driven pattern but is geographically centered on Germany and England rivalries and refereeing decisions involving players like Harry Kane.

C5 stands out as a specialized tactical blame community focused on national-team failure, specifically targeting Scotland and manager Steve Clarke with critiques of tactics and leadership.

Finally, C6 represents another controversy-oriented group, though it focuses on a different set of referee and goalkeeping incidents, frequently using terms like robbed and foul.

5.3.2 Sentiment related Insights

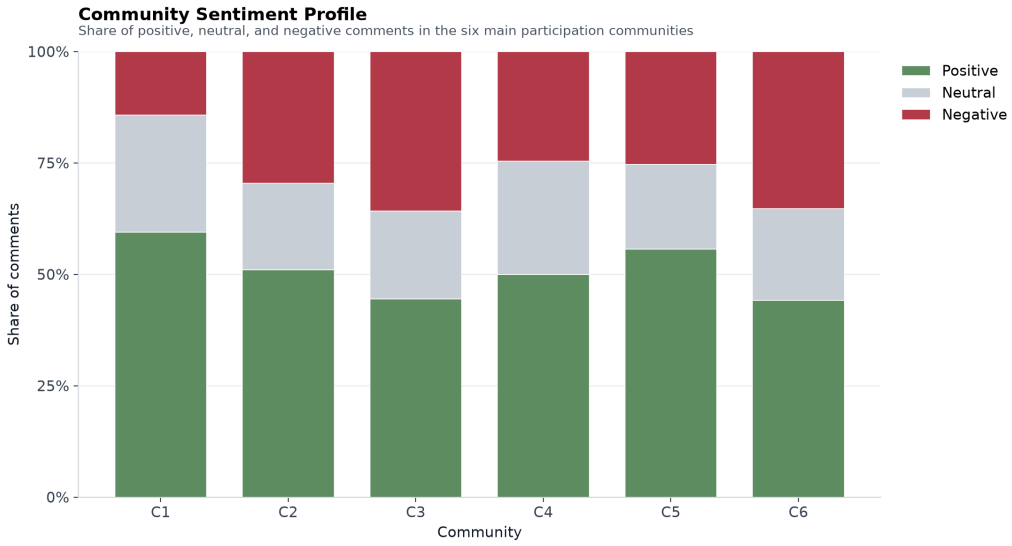


Figure 6: Sentiment per Main Community

The sentiment analysis reveals that while all participation communities exhibit a mix of polarities, significant variations emerge across the structural clusters. Communities C1 and C5 display a predominantly positive orientation, whereas C3 and C6 lean toward negative sentiment; C2 and C4 occupy a more neutral, intermediate position. These VADER-based scores provide critical supporting evidence, confirming that controversy-oriented communities maintain a more negative affective tone than those focused on match highlights or player appreciation.

5.3.3 Emotions related Insights

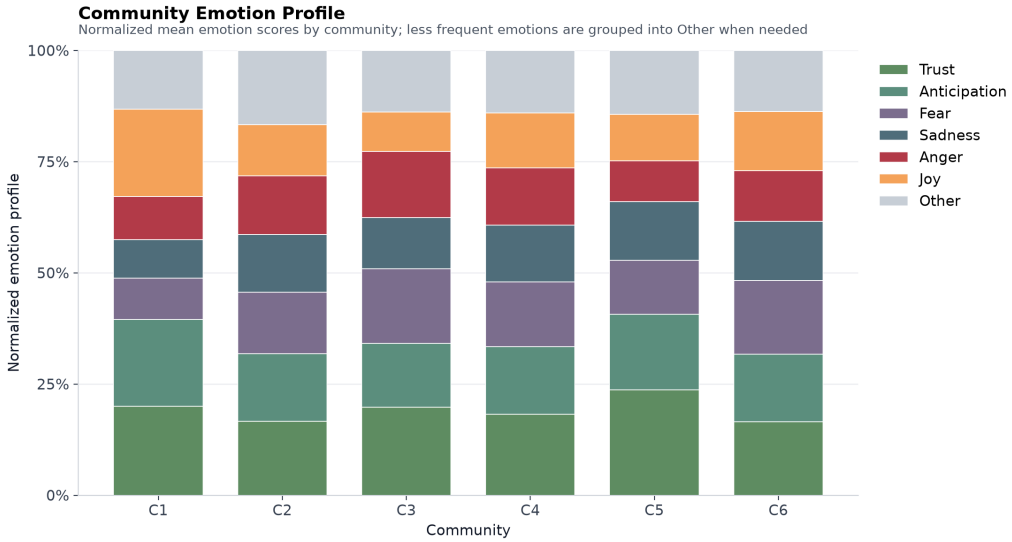


Figure 7: Emotions per Main Community

The emotional profiles extracted via NRCLex don't offer a sharply defined separation, showing relatively similar distributions across the network core. This lack of clear emotional fragmentation can be largely attributed to the nature of lexicon-based tools and the overlapping emotional vocabulary characteristic of global football discourse, where terms like great, trust, fear, and surprise appear in varying contexts across all groups. Despite this overlap, meaningful affective patterns are visible: C3 and C6 exhibit stronger components of anger, while C1 and C5 are more clearly defined by signals of joy and trust. Ultimately, while emotion profiles provide less structural distinction than TF-IDF keywords, they successfully validate the narrative divide between the anger-intensive discussions of match controversies and the positive affective language of celebratory fan communities.

5.3.4 NER related Insights

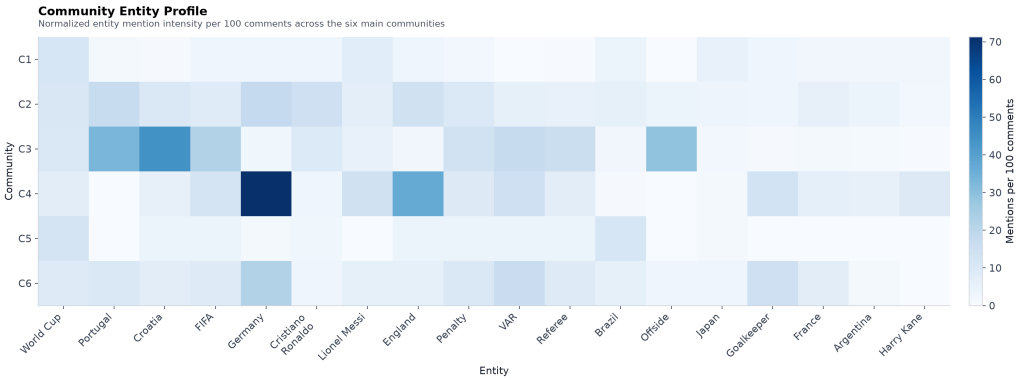


Figure 8: Entities per Main Community

The Named Entity Recognition (NER) and normalization process reveal that participation communities are distinguished not only by their emotional tone but, more fundamentally, by the specific narrative objects they center around. These entities—ranging from teams and star players to systemic concepts like VAR and officiating—act as the primary anchors for digital fan clustering. The distribution of normalized entities across the network core confirms that different communities engage with the tournament through distinct lenses, such as broad event appreciation, player-centric legacies, or concrete officiating disputes. Community C1 is characterized by a set of relatively generic entities, including World Cup, Japan, Argentina, and Lionel Messi. This profile aligns with the

interpretation of C1 as a broad, highlight-oriented community consisting of a diverse audience reacting to general tournament successes and high-visibility matches. In contrast, C2 exhibits a more concentrated focus on high-profile figures and teams involved in match drama, with frequent mentions of Portugal, Croatia, England, and Cristiano Ronaldo. The presence of the penalty entity alongside these figures supports the classification of C2 as a post-match reaction and star-player debate space, often centered on the intersection of individual performances and controversial game moments. The controversy-oriented communities, C3 and C6, are particularly well-defined by their entity profiles. C3 is dominated by terms such as Portugal, Croatia, Offside, VAR, and Referee, providing strong evidence of a community organized around a specific officiating dispute—namely, the offside decision in the Portugal-Croatia match. The entity distribution confirms that for this cluster, the narrative revolves entirely around the technicalities of the refereeing decision and the role of star players within that incident. Similarly, C6 centers on Germany, France, and entities like Goalkeeper and Foul, which, when triangulated with distinctive TF-IDF terms like robbed and keeper, points toward a narrative anchor focused on a separate goalkeeping or foul-related controversy. Community C4 presents a narrative intersection between national rivalries and systemic criticism, featuring prominent mentions of Germany, England, Harry Kane, and VAR. This suggests a discussion space organized around the specific refereeing incidents and post-match tensions inherent in high-level European team, such as Germany, unable to satisfy the expectations. Finally, while community C5 shows lower density for tournament-wide entities, its narrative focus is clearly established through localized entities such as Scotland, Steve Clarke, and Manager. This reinforces the interpretation of C5 as a specialized tactical blame community, where the discourse is primarily concerned with managerial accountability and national-team performance. Collectively, these entity-based profiles demonstrate that the structural separation of YouTube users is driven by their focus on different narrative subjects across the digital landscape of the World Cup.

5.3.5 Representative comments related Insights

The representative comments strongly support the interpretations suggested by TF-IDF, entities, sentiment, and emotions.

C1 has positive terms like: Messi, congratulations, japan, best, great. And the representative comments are very celebratory. They mention players scoring, ”what a day of football”, ”beautiful football”, ”huge love”, and congratulations to Argentina/Messi.

So C1 is not just large; it seems like a broad, positive, match-highlights/tournament-appreciation community.

C2 has: Portugal, England, Ronaldo, Croatia, Germany. The comments talk about FIFA favoritism, Portugal, Ronaldo, Croatia being robbed, penalties, and Germany squad debates. So C2 looks like a post-match debate community, with a mix of star-player discussion and controversy/fairness arguments.

C3 is extremely clear. Terms and entities include: ball, Croatia, touch, offside, Portugal, FIFA, Offside, Referee. The representative comments talk about whether the ball was touched, VAR, the referee, Croatia deserving a penalty, and claims that the decision was unfair or rigged.

C4 has: Germany, England, Germany, VAR, Goalkeeper, Harry Kane. The comments include arguments about whether a Germany incident was a foul, criticism of VAR, Germany losing identity, and comparisons to Kane/England decisions. So C4 seems like a Germany/England-related controversy and post-match debate community.

C5 is very clear: Scotland, Clarke, fans, scottish, Steve, Brazil, World Cup, Haiti. The comments discuss Steve Clarke, Scotland’s tactics, manager responsibility, and Scotland fans. This looks like a tactical blame/national-team criticism community.

C6 has: Robbed, right, Germany, VAR, Goalkeeper, Foul, Portugal. The comments talk about missed penalties, VAR, corrupt/incompetent officials, the Paraguay goalkeeper, simulation, and Germany being robbed. So C6 is another controversy community, probably focused on a different referee/goalkeeper/foul narrative than C3.

The communities are not just different in tone, they are organized around different narrative anchors.

The combination of these insights helps us to answer the second research question by transforming the structural partitions identified during community detection into substantive narrative spaces.

6 Bridge Analysis

The bridge analysis phase addresses the third research question by investigating the elements that connect different participation communities. While the previous chapters focused on the structural fragmentation of the network

and the specific narratives that define individual clusters, this analysis explores the users, videos, and entities that facilitate cross-community participation. The primary objective is to move from a view of isolated discussion spaces to a more integrated understanding of the World Cup 2026 YouTube landscape, identifying the specific anchors that act as shared focal points for a diverse and decentralized audience.

This analysis examines three distinct types of bridges within the community-assigned subset of the data. First, it identifies bridge users who occupy structural brokerage positions in the co-commenting network, measured through betweenness centrality. Second, it evaluates bridge videos that attract a balanced and diverse set of participants from multiple communities, using entropy-based metrics to measure the evenness of community representation. Finally, it detects bridge entities—such as specific teams, players, or controversial concepts like VAR—that appear across multiple narrative clusters, identified through a participation coefficient that distinguishes between community-specific and cross-community mentions.

By shifting the focus from internal community cohesion to external connectivity, the bridge analysis provides a final layer of interpretation to the study. It reveals how high-stakes events, global stars, and polarizing controversies can bridge narrative boundaries, helping to explain which elements of the tournament serve as common ground for the different communities. This step ensures that the project accounts not only for the diversity of fan discourse but also for the underlying connections that link different participation patterns together.

6.1 Bridge users

Bridge users are pseudonymized participants who occupy strategic brokerage positions within the user-user co-commenting network. To maintain methodological consistency, this analysis utilizes the same main graph defined during the community detection phase, restricted to edges with a weight of two or more. By focusing on this thresholded backbone, the bridge results remain aligned with the primary participation structure, ensuring that detected connectors are facilitating links between meaningful, repeated discussion spaces rather than accidental one-off overlaps.

The primary metric used to identify these actors is betweenness centrality, which measures the extent to which a node lies on the shortest paths between other nodes in the network. In the context of this study, a high betweenness score suggests that a user acts as a structural bridge, participating across different narrative communities and linking otherwise separated parts of the digital ecosystem. To provide a multi-dimensional view of these users, the analysis also incorporates degree (the number of unique co-commenting neighbors) and weighted degree (the total strength of shared participation ties), allowing for a distinction between high-volume participants and those who specifically occupy brokerage roles.

The calculation of bridge-user betweenness is performed as an unweighted shortest-path measure on the thresholded graph. This methodological choice is particularly defensible given the nature of the network: edge weights are utilized first to filter out noise and define the meaningful ties of the graph. Once the network is established, using unweighted betweenness avoids the common algorithmic conflict where larger weights, which indicate stronger social ties, are often misinterpreted as longer physical distances. Consequently, the analysis uses weights to define the strength and existence of connections while relying on the unweighted topology to identify structural brokerage.

It is critical to note that these metrics serve as structural indicators within the sampled participation network rather than evidence of real-world influence, friendship, or opinion leadership. Because the users remain pseudonymized and the network captures shared presence rather than direct interpersonal interaction, these bridge actors should be interpreted as individuals whose participation patterns happen to span multiple narrative clusters. This structural perspective allows the study to identify the human nodes that provide connectivity across the fragmented landscape of World Cup 2026 YouTube discourse.

6.2 Bridge videos

Bridge videos represent contents that attract participants from multiple narrative communities, functioning as shared focal points where diverse fan groups converge. Within the project design, these videos are treated as the primary catalysts for cross-community engagement, allowing for a thematic intersection of narratives even when users do not interact through direct replies. The purpose of this analysis is to move beyond the internal cohesion of individual clusters to identify the specific content that facilitates a more integrated digital landscape.

The identification of these hubs relies on a bridge-video score that combines community diversity with participation volume. For each video, the distribution of participating users across all detected communities is analyzed using normalized entropy. A high entropy score indicates that participation is evenly distributed across multiple communities, while a low score suggests that the discussion is dominated by a single narrative group. This diversity signal is then weighted by the log-transformed count of community-assigned users to ensure that top bridge videos represent both structurally diverse and substantively significant discussion hubs, without allowing very large videos to dominate the results solely through raw volume.

This analysis is intentionally conducted on the community-assigned subset of the data rather than the raw commenter dataset. Because bridge videos are defined specifically by their ability to link detected communities, only users who have been successfully integrated into the structural participation network are relevant to this calculation. Users falling outside the community-detection results provide no information regarding the connections between C1, C2, or other specific clusters. Therefore, the resulting bridge scores offer a precise view of connectivity within the detected community structure, highlighting the videos that effectively span the fragmented narrative boundaries of the World Cup 2026 YouTube ecosystem.

6.2.1 Entropy-based metric for Bridge Videos

Bridge videos are identified as those that act as shared focal points by attracting a diverse set of participants from across the different detected communities. The primary metric used for this evaluation is the bridge video score, which is designed to balance the evenness of community representation with the substantive volume of participation. To measure diversity, the analysis uses normalized community entropy, a score ranging from 0 to 1 where higher values indicate that participation is spread more evenly across multiple communities rather than being dominated by a single group. This diversity indicator is then multiplied by the natural logarithm of the total number of community-assigned users on the video, using the $log1p$ function to prevent high-volume videos from dominating the results solely based on raw counts. The resulting formula is:

$$\text{bridge video score} = \text{normalized community entropy} * log1p(\text{total community users on video})$$

. This approach ensures that the highest-ranking bridge videos are those that effectively integrate separate narrative clusters while maintaining a significant level of engagement.

6.3 Bridge entities

Bridge entities are normalized terms—such as teams, players, geopolitical entities, or controversy-specific concepts—that span multiple participation communities. While the preceding content analysis focused on the six largest clusters to interpret the most dominant narratives, the bridge analysis incorporates all detected communities. This inclusive scope is essential for identifying elements that connect the core network hubs to one another, as well as those that link the main discourse to smaller, peripheral groups in the network's long tail.

To ensure technical independence and granularity, bridge entities are recomputed directly from comment-level data rather than relying on compact community summaries. While community summary files remain valuable for interpreting the internal thematic focus of major clusters, the bridge analysis utilizes a dedicated workflow to track mentions across the entire community structure. This methodological shift from high-level salience to granular comment-level distribution allows for a more precise identification of entities that genuinely cross community boundaries, rather than those that are merely prominent within a single isolated group.

The extraction process utilizes the same normalized dictionary logic employed throughout the project, ensuring consistency across different analysis stages. Comments are linked to community assignments via pseudonymized author identifiers, and entities are extracted through case-insensitive matching. The normalization process maps various aliases to their canonical forms and prioritizes longer patterns during the matching process to ensure accuracy. Furthermore, multiple mentions of the same normalized entity within a single comment are deduplicated to prevent frequency bias, ensuring that each comment counts as a single unit of support for a given bridge entity.

In alignment with the broader bridging framework, this analysis is restricted to the community-assigned subset of the data. Comments whose authors fall outside the detected co-commenting structure are excluded, as they cannot provide meaningful evidence of connectivity between established communities. The resulting metrics prioritize the participation coefficient and total comment support, shifting the emphasis from an entity's popularity

within a single group to its structural distribution across the digital ecosystem. This approach provides a comprehensive map of narrative connectivity, identifying the anchors that unify the fragmented landscape of World Cup 2026 YouTube discourse regardless of their initial visibility in individual clusters.

6.3.1 Metrics for Bridge Entities

Bridge entities are evaluated through a multi-dimensional set of metrics, including the total number of communities involved, the count of main communities reached, a participation coefficient, and a final bridge-entity score. The metric for main community involvement indicates that an entity has appeared in at least one comment within the primary clusters, regardless of whether it ranks as a top or salient term in those specific groups. This allows the study to track the presence of specific narrative objects across the core of the network, even when their frequency within individual communities is relatively low.

The degree to which an entity is distributed across the network is captured by the participation coefficient (P). This metric measures whether an entity's mentions are concentrated within a single community or spread across multiple discussion spaces, and is calculated using the formula:

$$P(entity) = 1 - sum(p_c^2)$$

, where p_c represents the proportion of an entity's comment support attributed to community c. A coefficient of zero indicates that an entity is entirely specific to one community, while a higher value reflects a narrative object that successfully crosses community boundaries. For instance, an entity appearing only in C2 would have a participation coefficient of 0, whereas an entity like VAR, appearing evenly across three communities, would achieve a significantly higher coefficient of approximately 0.67, signaling a robust structural bridge.

To identify meaningful narrative connectors, the participation coefficient is integrated into a final bridge-entity score. This score is defined as:

$$\text{bridge entity score} = P(entity) * log1p(\text{total comments with entity})$$

. By combining the spread of an entity with the log-transformed total count of its mentions, the metric ensures that an entity must possess both broad community distribution and a baseline level of substantive support to be classified as a bridge. The use of the $log1p$ function is a deliberate methodological choice to prevent extremely frequent generic entities from dominating the rankings solely due to raw volume, while still filtering out entities that may appear in multiple groups but have negligible total presence.

The resulting scores allow for a clear distinction between broad contextual bridges and specific narrative bridges. High-ranking generic entities such as World Cup or FIFA often function as broad contextual anchors common to the entire tournament. In contrast, more specific entities—including VAR, Referee, Penalty, or high-profile figures like Cristiano Ronaldo—act as concrete narrative bridges that link diverse groups through shared focal points of controversy, match events, or star-player debates. It is important to emphasize that a high bridge-entity score reflects structural patterns of cross-community mention rather than any underlying agreement or shared stance among users; it identifies the common subjects of discourse that unify this digital landscape.

6.4 Why the Partecipation coefficient for entities and the Entropy-based metric for Videos

The use of the participation coefficient for entities and entropy for videos is not inconsistent, as both metrics share a similar underlying intuition while being applied to slightly different analytical contexts. For bridge entities, the primary objective is to determine whether an entity's comment support is concentrated within a single community or spread across structural boundaries; here, the participation coefficient is a natural choice for measuring cross-community presence. For bridge videos, the focus shifts to the diversity and balance of the audience within a specific discussion space, making entropy an ideal measure of evenness.

While the participation coefficient could technically be applied to videos with similar results, the normalized entropy metric specifically identifies discussion hubs with a balanced community distribution. A known characteristic of normalized entropy is that a video split evenly between two communities and one split between eight can receive similar scores; however, this is accounted for by the inclusion of the $log1p$ factor of total community users in the final formula. This ensures that the bridge video analysis prioritizes spaces that are not only diversely attended but also substantively significant in terms of participation volume. Ultimately, these methodological

choices ensure that the bridging metrics are tailored to the specific nature of the data being analyzed—entities as narrative anchors and videos as shared discussion environments.

6.5 Insights

6.5.1 Bridge Users Insights

The bridge analysis identifies a small but significant group of bridge users who occupy high-betweenness positions within the repeated co-commenting network. These participants serve as structural brokers between otherwise isolated narrative clusters. For instance, the top-ranking bridge user, a member of the VAR/controversy community (C3), exhibits a betweenness score of 0.2519 and a weighted degree of 94, having participated across 9 different video discussion spaces. This structural position indicates a high degree of cross-community participation; however, because these users are pseudonymized and their direct interactions are not observed, they must be interpreted strictly as structural connectors rather than social influencers or opinion leaders.

6.5.2 Bridge Video Insights

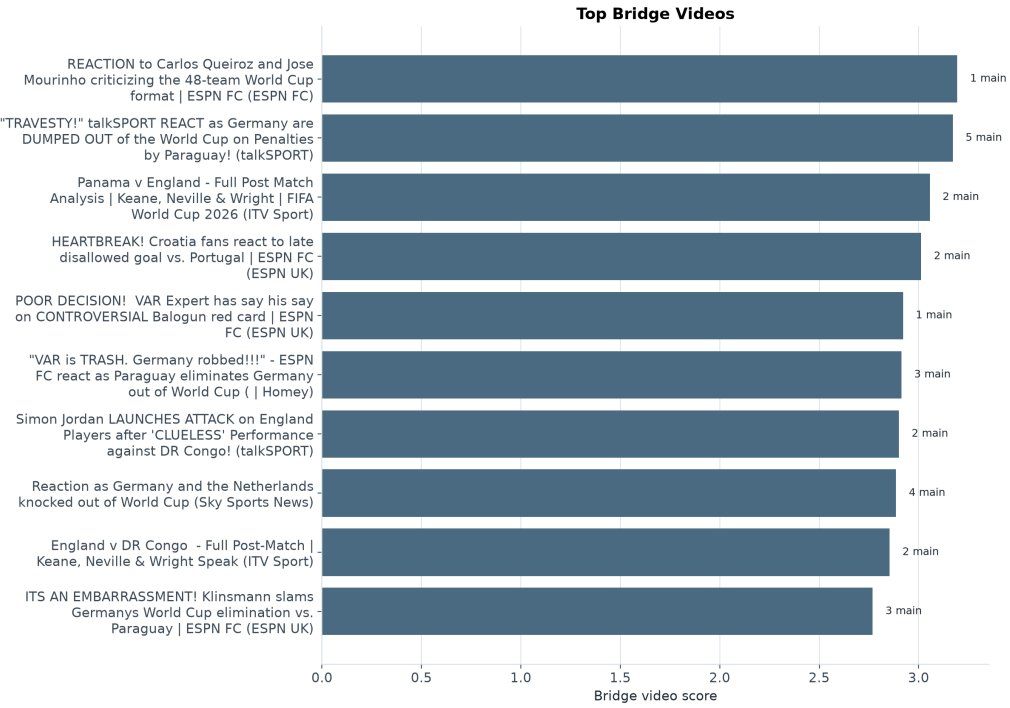


Figure 9: Main Bridge Videos

The bridge video analysis indicates that cross-community connectivity is primarily driven by three thematic catalysts: tournament eliminations, officiating controversies, and polarizing punditry. These videos function as shared discussion hubs where participants from diverse narrative communities converge, facilitating an overlap between otherwise fragmented segments of the network.

One of the most prominent examples of this bridging effect is found in the discourse surrounding Germany's elimination. The video titled "TRAVESTY! Germany dumped out by Paraguay" served as a significant cross-community event, attracting a diverse audience that included members of the Germany-oriented controversy group (C4), the post-match debate community (C2), the tactical blame cluster (C5), the match-highlights audience (C1), and additional controversy-focused users (C6). This distribution demonstrates that major tournament upsets act as broad focal points that transcend specific narrative boundaries, drawing together fans interested in tactical failure, refereeing complaints, and general match results.

Similarly, content centered on systemic officiating disputes serves as a structural link between different parts of the network. The video "VAR is TRASH. Germany robbed!!!" effectively bridged multiple controversy-oriented

communities (C6 and C4) alongside the broader post-match debate group (C2). In another instance, the video ”HEARTBREAK! Croatia fans react to late disallowed goal vs. Portugal” acted as a narrative bridge between the star-player and post-match debate community (C2) and the specialized VAR/offside controversy cluster (C3). These examples suggest that perceived injustices and technical officiating debates are uniquely capable of unifying users who otherwise occupy separate digital spaces.

Furthermore, pundit-reaction videos, particularly those focused on the England national team, emerge as consistent discussion hubs. Content such as ”Panama v England post-match analysis” and ”Simon Jordan attacks England players” frequently connects the post-match debate (C2) and controversy-oriented (C4) communities with various smaller peripheral groups. Ultimately, the data confirms that videos function as bridges when they combine high-stakes match outcomes, dramatic eliminations, or polarizing commentary, providing the common ground necessary to link the diverse participation patterns of the YouTube audience.

6.5.3 Bridge Entities Insights

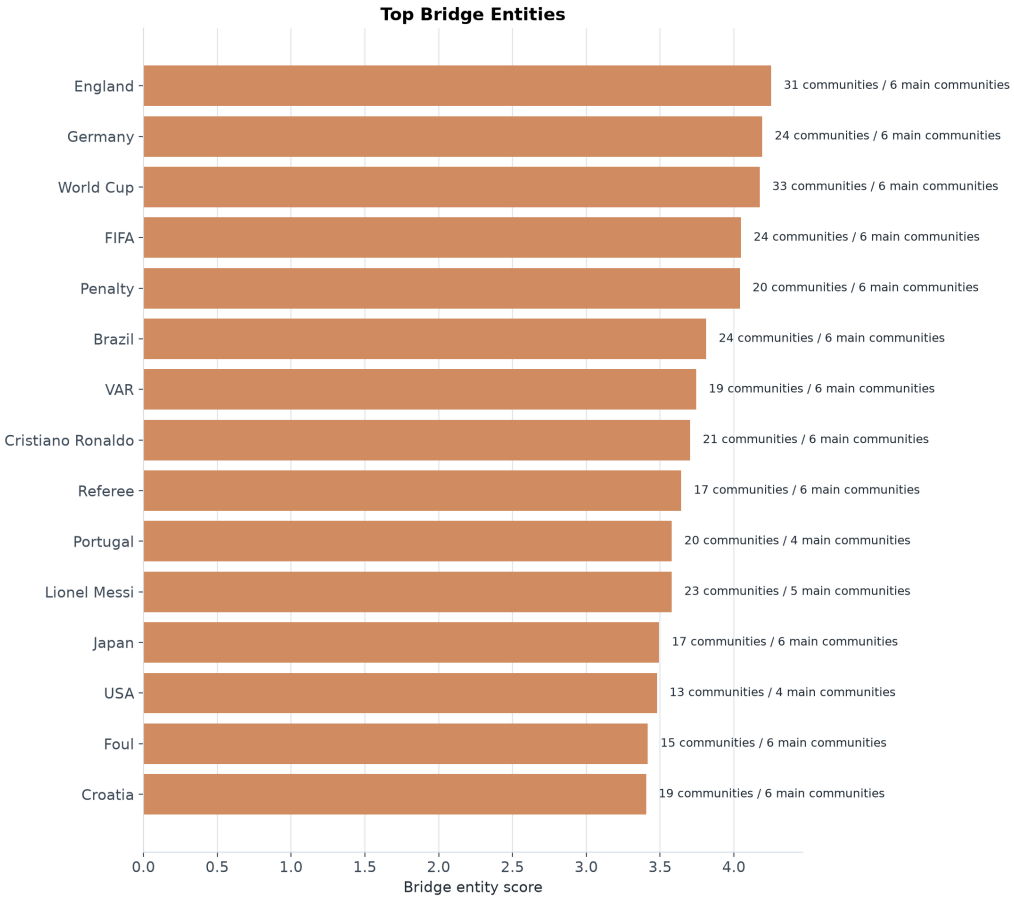


Figure 10: Main Bridge Entities

The analysis reveals that the most effective bridge entities are not characterized by frequency alone, but by their ability to combine significant comment support with a broad distribution across the participation communities. Within the dataset, England emerges as the primary bridge entity, followed closely by Germany, while World Cup, FIFA, Penalty, Brazil, VAR, Cristiano Ronaldo, Referee, and Portugal also rank among the top connectors. This distinction is critical for separating broad contextual bridges —generic anchors like World Cup and FIFA— from concrete narrative bridges. Entities such as England, Germany, VAR, and Referee function as narrative bridges because they are deeply embedded in match events, officiating controversies, and star-player debates that traverse multiple community boundaries. It is important to note that a high mention count does not automatically imply a bridging role; an entity that remains concentrated in a single cluster will have a low participation coefficient. Consequently, a high bridge-entity score signifies that a narrative object is both substantively discussed and structurally

distributed across the fragmented digital landscape.

7 Conclusions

In conclusion, this study demonstrates that the digital discourse surrounding the World Cup 2026 on YouTube is not a chaotic stream of comments, but a highly structured ecosystem of distinct participation communities organized around specific narrative anchors. By integrating structural, content, and connectivity analyses, we have mapped not only how fans are divided but also the specific elements that serve as common ground in a fragmented landscape.

The network analysis revealed that discussions aggregate into six primary communities, covering over 76% of the analyzed users. A modularity score of 0.5505 confirms that these clusters are meaningful participation structures rather than random groupings. While community detection provided the framework, content analysis defined the thematic identities of these clusters:

- **C1 (Highlights and Appreciation)**: The largest and most positive community, characterized by celebratory language and focused on match highlights and global icons like Lionel Messi.
- **C2 and C4 (Debate and Rivalry)**: Spaces defined by high-profile post-match reactions and national rivalries (notably England and Germany). These groups blend admiration for star players with heated arguments regarding match fairness.
- **C3 and C6 (VAR and Referee Controversy)**: Highly polarized communities centered on specific officiating incidents, such as disallowed goals. These clusters exhibit the highest levels of negative sentiment and anger-related emotions.
- **C5 (Tactical Blame)**: A specialized community focused on managerial accountability and tactical criticism, particularly regarding national team failures.

The bridge analysis identified the specific factors that integrate these otherwise isolated spaces, revealing how the decentralized YouTube landscape maintains a level of thematic cohesion.

- **Bridge Videos**: Content involving dramatic eliminations and pundit reactions acts as the strongest catalyst for cross-community engagement. For example, videos concerning Germany's unexpected exit functioned as major "intercommunity events," attracting users from highlight-oriented, tactical-blame, and controversy-driven clusters.
- **Bridge Entities**: While terms like "World Cup" serve as broad contextual bridges, specific entities such as "England," "Germany," "VAR," "Referee," and "Cristiano Ronaldo" function as concrete narrative bridges. These entities link the core network hubs to smaller peripheral groups, proving that certain polarizing themes transcend local or team-specific interests.
- **Bridge Users**: A small elite of users occupy strategic brokerage positions (high betweenness), facilitating structural connectivity between different narrative bubbles.

The findings confirm that dominant narratives in the World Cup 2026 digital ecosystem are not monolithic but are instead fragmented based on specific fan interests—ranging from celebratory appreciation to technical officiating critiques. However, these fragments are unified by high-stakes sporting drama and polarizing controversies. Ultimately, the data suggests that while fans may cluster into separate thematic groups, the shared need to negotiate the "truth" and "justice" of match events through collective debate remains a unifying force across the global YouTube audience.

8 Issues and possible future development

The findings of this study are subject to several methodological caveats that emphasize the need for a triangulated interpretation rather than viewing individual results in isolation. The co-commenting network and bridge analysis

identify patterns of shared participation and structural brokerage within specific discussion spaces, which should not be conflated with direct user interaction, interpersonal friendship, or real-world social influence. These structural results are inherently contingent upon platform-mediated factors —such as YouTube's relevance ranking— and methodological decisions, including the English-focused sampling, the 200-comment cap, and the choice of a weight-based edge threshold. Furthermore, community detection utilizes the Louvain heuristic to identify dense participation clusters; while these partitions are structurally meaningful, they represent one possible useful organization of the data rather than an objectively unique truth, and peripheral small communities or illustrative network layouts should not be overinterpreted. Finally, the content analysis relies on lexicon-based tools (VADER and NRCLex) and automated entity extraction that provide approximate signals of tone and subject matter but may fail to capture the nuances of sarcasm, football-specific slang, or complex irony. Because entity normalization is not linked to external knowledge bases and representative comments serve as illustrative rather than statistical proof, the final narrative labels must be understood as interpretations derived from the synthesis of video metadata, structural positions, and automated text metrics.

Beyond these specific analytical constraints, broader data collection limitations must be acknowledged to provide a full context for the findings. The dataset is not a statistically representative random sample of all World Cup 2026 YouTube discourse; instead, it is a narrative-oriented corpus heavily influenced by platform-mediated search results and ranking algorithms. Because the collection was restricted to top-level comments, it does not capture the complexity of full conversation threads or direct interpersonal interactions; therefore, co-commenting should be viewed strictly as an indicator of shared participation within a specific discussion space. Additionally, while the automatic filters and context labels used to categorize videos and comments are transparent, they are rule-based heuristics and remain imperfect. Finally, as the dataset is primarily English-focused, and certain categories or context types may be imbalanced, the analysis relies on normalized measures rather than raw counts to ensure a more robust comparison across the tournament's diverse narrative landscape.

For the future development of the project, several areas offer significant room for improvement to overcome current methodological simplifications. First, data collection could be expanded to include reply text and full conversations, allowing for a transition from a participation-only network to one based on direct interaction between users.

Regarding text analysis, the adoption of more advanced contextual language models (such as transformers) could drastically improve the detection of sarcasm, irony, and football-specific slang, surpassing the inherent limitations of lexicon-based methods like VADER and NRCLex. Furthermore, entity extraction could evolve toward true Entity Linking (NEL/NED) connected to external databases like Wikidata, providing a deeper categorization of narrative subjects.

Finally, extending the research to multiple languages and integrating qualitative influence metrics would overcome the current English-centric focus and better distinguish structural "bridge users" from genuine opinion leaders within the global community.